

Strategic Carbon Response: The Limit of a Direct Transfer

Q A Systematic Approach to Mechanism Design with Stochastic Dynamic Stability [arXiv:2608.29130](#)

P Emission-Forecasting-Based Spatial-Temporal Carbon Response: A Multi-Agent Attention-Enhanced Deep Learning Framework [arXiv:2607.26560](#)

Research phase: PAUSE after 1 round, 0 ITERATE, 0 REVISE. Edit phase: readable note

Abstract. Q studies payments that steer private resource requests towards a welfare optimum, while P forecasts carbon intensity and schedules flexible electricity demand. The thread asked whether Q’s mechanism could make strategic participants carry out P’s carbon-aware dispatch. The peers found a balanced charge for a restricted, fixed-forecast problem and a simple test for whether a predicted emissions reduction survives forecast error. Neither result covers P’s coupled, partly binary dispatch, and a conflict between two of Q’s printed conditions further weakens a direct transfer. The verifier therefore paused the thread pending a valid mechanism for P’s constraints and counterfactual emissions evidence.

1 Introduction

Q considers people or organisations that request shares of limited resources and know their own satisfaction better than the manager does. It proposes quadratic payments so that each participant’s best response leads to the allocation that maximises total satisfaction. Its stated results include a unique Nash equilibrium, budget balance at that equilibrium, voluntary participation, and convergence of a sample-based learning rule, each under stated assumptions. Its example concerns electric vehicle users choosing routes and charging stations [5].

P addresses a different problem in an electricity network. It forecasts the carbon intensity of power at each node for the next day, then schedules flexible loads, including mobile energy storage systems and distributed data centres, to reduce emissions. It tests the approach on a modified IEEE 33-bus network. The “agents” in its forecasting workflow clean inputs and respond to forecast errors; they are not modelled as self-interested owners of the scheduled loads [1].

The proposed connexion was to add private owners to P’s dispatch and use Q’s payments to make their choices follow a carbon-aware plan. This raised two questions. Could P’s decisions and constraints fit Q’s resource-allocation model? If they could, would a plan based on forecast carbon intensity still reduce emissions after the changed dispatch altered power flows?

2 What was done

The peers first compared the two optimisation problems. Q allows continuous requests in convex local sets, with a componentwise upper limit on their sum. P requires a workload balance across data centres and uses binary decisions for storage locations and travel. The peers therefore set aside a direct application to P’s full model. They explored a smaller model in which owners allocate splittable workload batches across destinations. That model can have convex local sets and destination capacities, but it changes who the agents are and what they control [5, 1].

Next, one peer proposed a carbon charge for a fixed forecast; the other checked its algebra. They then distinguished the forecast objective from emissions after dispatch, and derived an error test for comparing a proposed dispatch with a baseline. Finally, they checked Q’s printed payment and learning rules and compared the proposed connexion with earlier work. The verifier’s last check found an additional conflict between Q’s printed implementation and uniqueness conditions.

3 Results within the restricted model

Suppose there are $N > 1$ participants. Participant n chooses a nonnegative allocation vector x_n . Let $\hat{e}$ be a fixed vector of forecast carbon intensities and $\beta \geq 0$ the weight placed on forecast emissions. The peers checked the charge

$$\tau_n^C(x) = \beta \hat{e}^\top x_n - \frac{\beta}{N-1} \sum_{m \neq n} \hat{e}^\top x_m.$$

Here m indexes the other participants. Each forecast term appears once as a charge and $N-1$ times in rebates, so the charges sum to zero for every allocation. A participant's own marginal charge is $\beta \hat{e}$, because the rebate depends only on others' choices. Thus, if a valid version of Q's mechanism applies to this restricted allocation, adding the charge gives the same best responses as replacing each private satisfaction function $u_n(x_n)$ by $u_n(x_n) - \beta \hat{e}^\top x_n$. The resulting objective trades satisfaction against forecast emissions; it is not P's pure emissions objective. This is an algebraic construction, not an implementation result for P's dispatch.

The peers also obtained a comparison that does not depend on how a dispatch is chosen. Let $C(x)$ be emissions after power flows are recomputed for dispatch x , and let $\hat{C}(x)$ be its forecast value. For a proposed dispatch x^M and a baseline x^0 , suppose the absolute forecast error at *each* dispatch is at most δ . Then

$$C(x^0) - C(x^M) \geq \hat{C}(x^0) - \hat{C}(x^M) - 2\delta.$$

The argument is just the two error bounds: the baseline's realised emissions can be at most δ below its forecast, and the proposed dispatch's can be at most δ above. A forecast reduction greater than 2δ therefore certifies a realised reduction for those two decisions. P's historical forecast error does not supply this error bound for dispatches chosen under a new incentive scheme. Nor does maximising satisfaction less weighted forecast emissions, on its own, guarantee an emissions reduction.

4 Objections and unsettled points

The first obstacle is the agent model. P's balance equality links data centres, while Q's shared constraint is an upper capacity. Giving each workload batch its own choice over destinations could absorb the balance into local constraints, but it creates a new market. A mandatory batch also rules out the zero allocation used in Q's voluntary-participation argument. The nonnegative rebate in the charge above is therefore insufficient to establish participation without a feasible outside option. P's binary storage routes remain outside that restricted model [5, 1].

The second obstacle is the carbon signal. P's carbon-flow calculation makes nodal intensity depend on dispatch, whereas its scheduling objective uses forecast intensities as fixed coefficients. A fixed charge can match the forecast objective but need not match the marginal effect on realised emissions. The peers' illustrative two-location example showed that a forecast can favour one location even when moving all demand there makes realised emissions larger. This example was a logical check, not a fit to P's network. P describes a closed-loop assessment and reports a 33.86% reduction between its Cases 2 and 3; the supplied text does not establish whether that result survives private costs and counterfactual forecast error [1]. The peers did not infer that P's reported comparison was wrong.

Checks of Q raised separate cautions. The reduced payment printed beside its learning update has an own-price quadratic sign that differs from its full payment; a peer expanded both expressions and recorded corrected blocks. Q's convergence theorem requires a growing sample size, but its numerical schedule gives one sample at every iteration. These discrepancies call for correction and verification before transferring its learning claim; they do not show that its plotted trajectories fail [5].

The verifier identified a more direct conflict in Q's printed conditions. Let Ψ be Q's matrix of payment curvature, let K be the number of resource coordinates, and let A_{nm}^n denote its price-to-price block for participant n and participant m . Q's implementation condition P2(i) says $\sum_m A_{nm}^n = 0$. For any nonzero K -vector p , take a direction v that changes every participant's proposed price by p and leaves every resource request unchanged. The price blocks then give $v^\top(\Psi + \Psi^\top)v = 0$. Yet Q's uniqueness condition P1(ii) requires this quantity to be at most $-v^\top v < 0$, for a positive constant v . As printed, these two conditions cannot both hold. This does not rule out a corrected mechanism, but the thread contains no such correction or replacement proof [5].

The prior-work check also narrowed any novelty claim. Earlier work covers incentives for spatial data-centre demand response [3], marginal emissions signals for demand response [6], and carbon-aware

scheduling under uncertain grid signals [7]. It also covers equilibrium effects of average carbon signals [8], joint electricity-carbon pricing with budget balance [4], and carbon-flow-aware demand response [2]. The peers did not establish that the proposed combination is absent from this literature. The balanced charge and the 2δ inequality are useful checks, but neither is a new result for P's full setting.

5 Why the thread stopped

The verifier paused the thread because the conditional charge and error test do not connect Q's stated mechanism to P's coupled, partly binary dispatch; the prior-work check also left novelty unproved. The conflict between Q's P2(i) and P1(ii) makes a direct appeal to its printed conditions untenable. The smallest useful restart is a worked restricted data-centre case: specify the strategic owners and their outside option, repair and verify the needed Q conditions or supply another mechanism for the balance constraint, and measure forecast and realised emissions for its dispatch and a comparator on P's network. A claim about the full storage and data-centre system would then still need to handle its routing decisions.

References

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